

to the statistic 't' is standard normal i.e., viz., $Z = \frac{obsv.Value - Exp.Value}{Std Error}$ or

$$Z = \frac{t - E(t)}{S.E(t)} \sim N(0, 1), \text{ as } n \rightarrow \infty. \text{ The value of } Z \text{ given by the above relation}$$

under the null hypothesis is known as the 'test statistic'. The critical value of test statistic at level of significance α for a two-tailed-test is given by z_α where z_α is determined by the equation $P[|Z| > z_\alpha] = \alpha$ i.e., z_α is the value so that the total area of the critical region on both tails is α . Since the normal probability curve is symmetrical curve, from $P[|Z| > z_\alpha] = \alpha$ we can write,

$$P[Z > z_\alpha] + P[Z < -z_\alpha] = \alpha$$

$$\Rightarrow P[Z > z_\alpha] + P[Z < z_\alpha] = \alpha$$

$$\Rightarrow 2P[Z > z_\alpha] = \alpha$$

$$\Rightarrow P[Z > z_\alpha] = \alpha/2$$

i.e., the area of each tail is $\alpha/2$.

Thus, z_α is the value such that the area to the right of z_α is $\alpha/2$ and to the left of $-z_\alpha$ is $\alpha/2$.

In case of single-tail alternative, the critical value z_α is determined so that the total area to the right of it (for right-tailed test) is α and for left-tailed test the total area to the left of $-z_\alpha$ is α , i.e., $P[Z > z_\alpha] = \alpha$ (for right-tailed test) and $P[Z < -z_\alpha] = \alpha$ (for left tailed test).

Thus, the significant or critical value of Z for a single-tailed test (left or right) at the level of significance ' α ' is the same as the critical value of Z for a two-tailed test at level of significance ' 2α '. The critical values of Z at commonly used levels of significance for both two-tailed and single-tailed test are given in Table 31.1:

Table 31.1: Critical Values of Z (commonly used)

Critical Values (z_α)	Level of Significance (α)		
	1%	5%	10%
Two-tailed test	$ Z_\alpha = 2.58$	$ Z_\alpha = 1.96$	$ Z_\alpha = 1.645$
Right-tailed test	$Z_\alpha = 2.33$	$Z_\alpha = 1.645$	$Z_\alpha = 1.28$
Left-tailed test	$Z_\alpha = -2.33$	$Z_\alpha = -1.645$	$Z_\alpha = -1.28$

Note: If n is small (usually less than 30), then the sampling distribution of the test statistic Z will not be normal and in that case we cannot use the above significant values, which have been obtained from normal probability curves.

31.3 CRITICAL REGION METHOD & P-VALUE METHOD OF HYPOTHESIS TESTING

In the Figures 31.1, 31.2 and 31.3, the dividing lines of the test criteria (i.e., the boundaries of critical regions, or the critical values) require the knowledge of z_{α} or $z_{\alpha/2}$. These values are provided above for the different levels of significance (for one tail test or two tail test) or in the table of Standard Normal Distribution. Similarly, if the sample distribution of the test statistic is based on t distribution, or a Chi Square distribution or a F distribution the corresponding tables will provide the necessary values for the respective values of α . It is rather convention to base the statistical hypothesis testing on the level of significance $\alpha = 0.05$ or $\alpha = 0.01$ or atmost $\alpha = 0.10$.

Apart from the **Critical region method (Rejection Region method)** of hypothesis testing (discussed above) there is another method to test the hypothesis called the P value method.

With reference to a test statistic, the P value (or the probability value) is presented by the shaded region as in Fig. 31.4. The P value is compared with α value (instead of comparing the test statistics with the tabulated value). We reject the Null hypothesis if the shaded region is less than the α value.

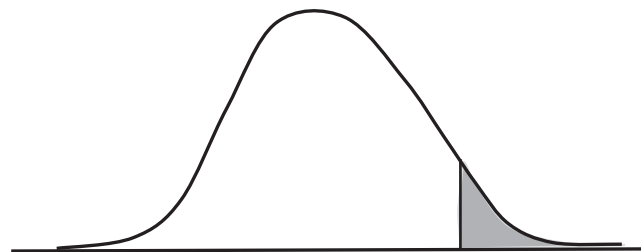


Fig. 31.4: Diagram for Definition of P-Values

Thus, corresponding to an observed value of a test statistic the P value is the lowest level of significance at which the Null hypothesis could be rejected.

Steps for computing the P-value for a two tailed test:

- 1) Compute the value of test statistic (say Z)
- 2) If the value of the test statistic is in upper tail ($Z > 0$), find the area under the standard normal curve (i.e., probability to the right of Z). If the value of the test statistic is in the lower tail ($Z < 0$), find the area under the standard normal curve to the left of Z.
- 3) Double the tail area, or probability, obtained in step 2 to obtain the P-value (since we are considering a two-tailed test).
- 4) a) If the P value $< \alpha \Rightarrow$ Reject the Null Hypothesis.
b) If the P value $\geq \alpha \Rightarrow$ Do not reject the Null Hypothesis

Guidelines for P values:

- 1) If **P value** ≤ 0.01 , Reject the H_0 . The difference is significant (1%).
- 2) If $0.01 < \text{P value} \leq 0.05$, Reject the H_0 . The difference is significant (5%).
- 3) If $0.05 < \text{P value} \leq 0.10$, Reject the H_0 . Consider the consequences of Type I Error before rejecting H_0 .
- 4) If P value is > 0.10 , do not reject the H_0 . The difference is not significant.

Procedure for Testing of Hypothesis: We now summarise below the various steps in the testing of a statistical hypothesis in a systematic manner.

- 1) **Null Hypothesis:** Set up the null hypothesis H_0 .
- 2) **Alternative Hypothesis:** Set up the alternative hypothesis H_1 . This will enable us to decide whether we have to use a single-tailed (right or left) test or two-tailed test.
- 3) **Level of Significance:** Choose the appropriate level of significance (α) depending on the reliability of the estimates and permissible risk. This is to be decided before the sample is drawn, i.e., α is fixed in advance.
- 4) **Test Statistic (or Test Criterion):** Compute the test statistic $z = \frac{t - E(t)}{S.E(t)}$ under the null hypothesis.
- 5) **Conclusion:** We compare z , the computed value of Z in step 4 with the significant value (tabulated value) z_α , at the given level of significance ' α '.

If $|Z| < z_\alpha$, i.e., if the calculated value of Z (in modulus) is less than z_α , we say it is not significant. By this, we mean that the difference $t - E(t)$ is just due to fluctuations of sampling and the sample data do not provide us sufficient evidence against the null hypothesis which may therefore be accepted.

If $|Z| > z_\alpha$, i.e., if the calculated value of Z is greater than the critical or significant value, then we say that it is significant and the null hypothesis is rejected at level of significance ' α ', i.e., with confidence coefficient $(1 - \alpha)$.

Table 31.2: Summary of Hypothesis Tests about Population Mean (σ Known)

	Lower Tail Test	Upper Tail Test	Two-Tailed Test
Hypotheses	$H_0 : \mu \geq \mu_0$ $H_a : \mu < \mu_0$	$H_0 : \mu \leq \mu_0$ $H_a : \mu > \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$
Test Statistic	$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$	$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$	$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$
Rejection Rule: p-value Approach	Reject H_0 if $\text{p-value} \leq \alpha$	Reject H_0 if $\text{p-value} \leq \alpha$	Reject H_0 if $\text{p-value} \leq \alpha$

Rejection Rule: Critical Value Approach	Reject H_0 if $z \leq -z_{\alpha}$	Reject H_0 if $z \geq z_{\alpha}$	Reject H_0 if $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$
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Example: A researcher reports that the average salary of Junior manager is more than Rs. 42000/. A sample of 30 junior manager has a monthly mean salary Rs. 43260/. At $\alpha = 0.05$, test the claim that the junior manager earns more than Rs 42000/. The SD of population is Rs 5260.

Ans. Given $n = 30$, $\bar{x} = \text{Rs } 43260$, $\mu = \text{Rs } 42000$, $\sigma = \text{Rs } 5260$, $\alpha = 0.05$

Let, $H_0 : \mu \leq \text{Rs } 42000$ and $H_1 : \mu > \text{Rs } 42000$ (Thus it is therefore a Right tail test.)

The critical z_{α} values for $\alpha = 0.05$ is 1.65

The test statistics is: $z = \frac{\text{obsvV} - \text{Exp V}}{\text{Std Error}} = \frac{(\bar{x} - \mu)}{\sigma / \sqrt{n}} = \frac{43260 - 42000}{5260 / \sqrt{30}} = 1.32$

Since $z = 1.32 < z_{\alpha} = 1.65$, the test statistic is not lying in the critical region $\Rightarrow$ do not Reject H_0 . There is not enough evidence to support the claim that the Junior Manager earns more than Rs 42000/ on an average.

Though the sample mean is higher than the population mean, it is not significantly higher. Hence the difference may be due to chance.

Example: A researcher claims that the average wind speed of a certain city is 8 miles/hr. A sample of 32 days shows an average wind speed of 8.2 miles/hr. The standard deviation of the sample is 0.6 mile/hr. At $\alpha = 0.05$, is there enough evidence to reject the claim? Use P value method.

Ans. Given $n = 32$, $\bar{x} = 8.2$, $\mu = 8$, $s = 0.6$, $\alpha = 0.05$

Let, $H_0 : \mu = 8$ (Claim) and $H_1 : \mu \neq 8$.

Computing the test statistics:

$$z = \frac{\text{ObsvV} - \text{Exp V}}{\text{Std Error}} = \frac{(\bar{x} - \mu)}{\sigma / \sqrt{n}} = \frac{8.2 - 8}{0.6 / \sqrt{32}} = 1.89$$

Using the z table the area corresponding to $z = 1.89$ is 0.4706

For deriving the P value, we need to deduct 0.4706 from 0.5

$$\Rightarrow 0.5 - 0.4706 = 0.0294.$$

It is a two-tailed test, $0.0294 > \frac{\alpha}{2} = 0.025 \Rightarrow$ P value is greater than $\frac{\alpha}{2}$.

$\Rightarrow$ Accept the Null Hypothesis

Thus, there is not enough evidence to reject the claim that the average wind speed is 8 miles/hr.

Check Your Progress 1

- 1) What is meant by test of significance?
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- 2) State the purpose of hypothesis testing.
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.....
- 3) Define: i) Critical Region and Level of Significance & Confidence Limit,
ii) Two Tail Test & One Tail Test.
.....
.....
- 4) Suppose that 100 tires made by a certain manufacturer lasted on the
average 21,819 miles with a standard deviation of 1,295 miles. Test the
Null hypothesis $\mu = 22000$ miles against the alternative hypothesis
 $\mu \neq 22000$ miles at the 0.05 levels of significance. Check the result with the
P value method as well.
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31.4 TYPE I AND TYPE II ERROR

The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In practice, we decide to accept or reject the lot after examining a sample from the population. As such, we are liable to commit the following two types of errors:

Type I Error: Reject H_0 when it is true.

Type II Error: Accept H_0 when it is wrong, i.e., accept H_0 when H_1 is true.

If we write, $P[\text{Reject } H_0 \text{ when it is true}] = P[\text{Reject } H_0 | H_0] = \alpha$

and $P[\text{Accept } H_0 \text{ when it is wrong}] = P[\text{Accept } H_0 | H_1] = \beta$

then α and β are called the sizes of type I error and type II error respectively.

In practice, type I error amounts to rejecting a lot when it is good and type II error may be regarded as accepting the lot when it is bad.

Thus, $P[\text{Rejecting a lot when it is good}] = \alpha$

and $P[\text{Accepting a lot when it is bad}] = \beta$

where α and β are referred to as 'Producer's risk' and 'Consumer's risk' respectively.

The probability of type I error is necessary for constructing a test of significance. It is in fact the 'size of the critical region'. The probability of type II error is used to measure the 'power' of the test in detecting the falsity of the null hypothesis.

It is desirable that the test procedure be so framed that it minimises both the types of error. But this is not possible, because for a given sample size, an attempt to reduce one type of error is generally accompanied by an increase in the other type. The test of significance is usually designed as to limit the probability of type I error to a specified value (usually 5% or 1%) and at the same time to minimize the probability of type II error. Note that when the population has a continuous distribution,

Probability of type I error = Level of significance = Size of critical region

31.5 POWER OF THE TEST

The null hypothesis is accepted when the observed value of test statistic lies outside the critical region as determined by the test procedure. Type II error is committed when alternative hypothesis holds even though null hypothesis is not rejected, i.e., the test statistic lies outside the critical region. Hence, the probability of type II error is a function of the value for which alternative hypothesis holds.

If β is the probability of type II error (i.e., probability of accepting H_0 when H_0 is false), then $(1 - \beta)$ is called the 'power function' of the test hypothesis H_0 against the alternative hypothesis H_1 . The value of the power function at a parameter point is called the 'power of the test' at that point.

Power = $1 - \text{Probability of Type II error} = \text{Probability of rejecting } H_0 \text{ when } H_1 \text{ is true.}$

31.6 OPTIMUM TEST UNDER DIFFERENT SITUATIONS

The discussion till now enabled us to obtain the best test under different situations. In any testing problem, the first two steps, viz., (i) the form of the population distribution with the parameter(s) of interest and (ii) the framing of H_0 and H_1 are obvious from the very description of the problem. The most crucial step is the choice of the 'best test', i.e., the basic statistic 't' and the critical region W. By best test we mean the one in which besides controlling α at any desired low level, we also have the minimum type II error β or maximum power $(1 - \beta)$. This is when compared to the β of all other tests having the same α . This leads to the following definition.

31.6.1 Most Powerful Test

Let us consider the problem of testing a simple hypothesis $H_0: \theta = \theta_0$ against a simple alternative hypothesis $H_1: \theta \neq \theta_0$.

Definition: The critical region W is the most powerful critical region of size α (and the corresponding test the most powerful test of level α) for testing $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ if $P(\mathbf{x} \in W | H_0) = \alpha$ and $P(\mathbf{x} \in W | H_1) \geq P(\mathbf{x} \in W_1 | H_1)$ for every other critical region W_1 satisfying the first condition.

31.6.2 Uniformly Most Powerful Test

Let us now take up the case of testing a simple null hypothesis $H_0: \theta = \theta_0$ against a composite alternative hypothesis $H_1: \theta \neq \theta_0$. In such a case, for a predetermined α , the best test for H_0 is called the uniformly most powerful test of level α .

Definition: The critical region W is called the uniformly most powerful critical region of size α (and the corresponding test as uniformly most powerful test of level α) for testing $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ if $P(\mathbf{x} \in W | H_0) = \alpha$ and $P(\mathbf{x} \in W | H_1) \geq P(\mathbf{x} \in W_1 | H_1)$ for all $\theta \neq \theta_0$, whatever the other critical region W_1 satisfying the first condition may be.

31.7 TEST PROCEDURE UNDER NORMALITY ASSUMPTION

The general procedure to be followed in testing a statistical hypothesis has been explained in the previous sections. We shall now take up one by one some of the common tests that are made on the assumption of normality for the underlying random variable or variables.

31.7.1 Problems Regarding the Univariate Normal Distribution

Consider a population where x is normally distributed with mean μ and standard deviation σ . Let $x_1, x_2, \dots, x_n$ be a random sample obtained from this distribution.

We shall denote by $\bar{x}$ the sample mean of x : $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and by s'^2 the sample

variance of x : $s'^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. The distinction between s^2 and s'^2 is to be

noted. In s'^2 the divisor is $(n-1)$, which makes it an unbiased estimator of σ^2 .

$$\text{For } \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - n(\bar{x} - \mu)^2,$$

$$\text{so that } E(s'^2) = \frac{1}{n-1} E\left\{\sum_{i=1}^n (x_i - \bar{x})^2\right\} = \frac{1}{n-1} E\left\{\sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2\right\}$$

$$= \frac{1}{n-1} \left\{ \sum_i \text{var}(x_i) - n \text{var}(\bar{x}) \right\} = \frac{1}{n-1} \left\{ n\sigma^2 - n \frac{\sigma^2}{n} \right\} = \sigma^2.$$

Case I: μ unknown, σ known

Here we may be required to test the null hypothesis $H_0: \mu = \mu_0$. It has already been shown that the test procedure for H_0 in this case is based on the statistic,

$\sqrt{n}(\bar{x} - \mu_0)/\sigma$, which is distributed as a normal variate (τ) under this hypothesis.

- For the alternative $H: \mu > \mu_0$, H_0 is rejected if for the given sample $\tau > \tau_a$ (and is accepted otherwise)
- For the alternative $H: \mu < \mu_0$, H_0 is rejected if for the given sample $\tau < \tau_{1-a} (= -\tau_a)$.
- For the alternative $H: \mu \neq \mu_0$, H_0 is rejected if for the given sample $|\tau| > \tau_{a/2}$.

In each case, α denotes the chosen level of significance.

As regards the problem of interval estimation of μ , it has been shown that the limits $(\bar{x} - \tau_{a/2} \cdot \sigma / \sqrt{n})$ and $(\bar{x} + \tau_{a/2} \cdot \sigma / \sqrt{n})$, computed for the given sample, are the confidence limits for μ with confidence coefficient $(1 - \alpha)$.

Case II: μ known, σ unknown

Here one may be interested in testing a hypothesis regarding σ or in estimating σ .

A sufficient statistic for σ is $\sum_i (x_i - \mu)^2$ or, $\sqrt{\sum_i (x_i - \mu)^2}$. It is seen that such an x_i is a normal variable with mean μ and standard deviation σ .

Hence, $\sum_i \left(\frac{x_i - \mu}{\sigma} \right)^2$, being the sum of squares of n independent normal deviates, is distributed as χ^2 with $df = n$.

For testing $H_0: \sigma = \sigma_0$, we make use of the fact

$$\sum_i \left(\frac{x_i - \mu}{\sigma} \right)^2 = \sum_i (x_i - \mu)^2 / \sigma_0^2 \text{ is a } \chi^2 \text{ with } df = n \text{ under this hypothesis.}$$

- For the alternative $H: \sigma > \sigma_0$, H_0 is rejected in case for the given sample $\chi^2 > \chi_{a,n}^2$.
- For the alternative $H: \sigma < \sigma_0$, H_0 is rejected if for the given sample $\chi^2 < \chi_{1-a,n}^2$.
- For the alternative $H: \sigma \neq \sigma_0$, H_0 is rejected if for the given sample $\chi^2 < \chi_{1-a/2,n}^2$ or, $\chi^2 > \chi_{a/2,n}^2$.

As consistent (but biased) point estimate of σ , we have $\sqrt{\frac{1}{n} \sum_i (X_i - \mu)^2}$. To get a confidence interval of σ , we note that

$$P \left[\chi_{1-a/2,n}^2 \leq \sum_i \frac{(x_i - \mu)^2}{\sigma^2} \leq \chi_{a/2,n}^2 \right] = 1 - \alpha$$

$$\text{or, } P \left[\sum_i \frac{(x_i - \mu)^2}{\chi_{a/2, n}^2} \leq \sigma^2 \leq \sum_i \frac{(x_i - \mu)^2}{\chi_{1-a/2, n}^2} \right] = 1 - \alpha$$

$$\text{or, } P \left[\sum_i \frac{(x_i - \mu)^2}{\chi_{a/2, n}^2} \leq \sigma^2 \leq \sum_i \frac{(x_i - \mu)^2}{\chi_{1-a/2, n}^2} \right] = 1 - \alpha$$

The confidence limits of σ^2 with confidence coefficient $(1 - \alpha)$ are therefore, $\frac{(x_i - \mu)^2}{\chi_{a/2, n}^2}$ and $\frac{(x_i - \mu)^2}{\chi_{1-a/2, n}^2}$. The confidence limits of σ are just the positive square roots of these quantities, with the same confidence coefficient, $(1 - \alpha)$.

Case III: μ and σ both unknown

In this case, $\bar{x}$ and s' are jointly sufficient for μ and σ . Here to test $H_0: \mu = \mu_0$ or to have confidence limits for μ , one cannot use the statistic $\sqrt{n}(\bar{x} - \mu)/\sigma$ since σ is unknown. σ in this case is replaced by its sample estimate, $s' = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$. The resulting expression will be $\sqrt{n}(\bar{x} - \mu)/s'$.

Now from the discussion made in the last unit, it is clear that

$$\frac{(n-1)s'^2}{\sigma^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2} \text{ is a } \chi^2 \text{ with df} = (n-1) \text{ and is distributed independently of } \bar{x}.$$

Thus, $\sqrt{n}(\bar{x} - \mu)/s' = \frac{\sqrt{n}(\bar{x} - \mu)/\sigma}{\sqrt{\frac{(n-1)s'^2}{\sigma^2} / (n-1)}}$, being of the form $\frac{\tau}{\sqrt{\chi^2/(n-1)}}$,

where χ^2 has df = $(n-1)$ and is independent of τ , is distributed as a 't' with

$$\text{Df} = (n-1)$$

To test $H_0: \mu = \mu_0$ we may, therefore use the statistic $t = \sqrt{n}(\bar{x} - \mu_0)/s'$ with df = $(n-1)$. We shall have to compute 't' (computed from the given sample) with $t_{\alpha, n-1}$ or 't' with $-t_{\alpha, n-1}$ or $|t|$ with $t_{\alpha/2, n-1}$, according as the alternative of interest is $H: \mu > \mu_0$, $H: \mu < \mu_0$ or $H: \mu \neq \mu_0$.

In order to obtain confidence limits for μ , we see that

$$P \left[(\bar{x} - t_{a/2, n-1}) \frac{s'}{\sqrt{n}} \leq \mu \leq (\bar{x} + t_{a/2, n-1}) \frac{s'}{\sqrt{n}} \right] = 1 - \alpha$$

The $100(1 - \alpha)\%$ confidence limits for μ will, therefore, be $(\bar{x} - t_{a/2, n-1}) \frac{s'}{\sqrt{n}}$ and

$$(\bar{x} + t_{a/2, n-1}) \frac{s'}{\sqrt{n}}, \text{ these having been computed from the given sample.}$$

The above procedure is called as the *Student's t-test*.

In this case, we may have also the problem of testing $H_0: \sigma = \sigma_0$ or the problem of obtaining confidence limits for σ . From what has been said above, it is clear that

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma_0^2} = \frac{(n-1) s'^2}{\sigma_0^2}$$
 is, under the hypothesis H_0 , a χ^2 with $df = (n-1)$. This

provides us with the test for H_0 . The value of this χ^2 , computed from the given sample, is compared with $\chi_{a,n-1}^2$ or $\chi_{1-a,n-1}^2$, according as the alternative is

$H: \sigma > \sigma_0$ or $H: \sigma < \sigma_0$.

For the alternative $H: \sigma \neq \sigma_0$, on the other hand, the computed value is to be compared with both $\chi_{1-a/2,n-1}^2$ and $\chi_{a/2,n-1}^2$, H_0 being rejected if the computed value is smaller than the former or exceeds the latter value.

Since $P\left[\chi_{1-a/2,n-1}^2 \leq \frac{(n-1) s'^2}{\sigma^2} \leq \chi_{a/2,n-1}^2\right] = 1 - \alpha$

i.e., $P\left[\frac{(n-1) s'^2}{\chi_{a/2,n-1}^2} \leq \sigma^2 \leq \frac{(n-1) s'^2}{\chi_{1-a/2,n-1}^2}\right] = 1 - \alpha$

The confidence limits for σ^2 are therefore, $\frac{(n-1) s'^2}{\chi_{a/2,n-1}^2}$ and $\frac{(n-1) s'^2}{\chi_{1-a/2,n-1}^2}$. The confidence limits with the same confidence coefficient $(1 - \alpha)$ to σ are, of course, the positive square roots of these quantities.

Check Your Progress 2

- 1) Which is wider, a 95% or a 99% confidence interval?
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.....
.....
- 2) When you construct a 95% confidence interval, what are you 95% confident about?
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.....
.....
.....
- 3) When computing a confidence interval, when do you use t and when do you use z?
.....
.....
.....